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### Semiconductor structure with an air gap, method for forming same, and stacked structure

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#### Abstract

A semiconductor structure includes: a base, including a substrate and a dielectric layer, the substrate being provided with a first front surface and a first back surface which are opposite, and the dielectric layer being located at the first front surface; a connecting hole, penetrating through the substrate and extending into the dielectric layer; a first insulating layer, located at the surface of an inner wall of the connecting hole; a protective barrier layer, located at the surface of the first insulating layer and grounded; a second insulating layer, located at the surface of the protective barrier layer; and a connecting structure, located at the surface of the second insulating layer and filling up the connecting hole.

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| <b>Inventors:</b> | <b>Wang; Luguang (Hefei, CN)</b>                      |
| <b>Applicant:</b> | <b>CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)</b> |
| <b>Family ID:</b> | <b>1000008819494</b>                                  |
| <b>Assignee:</b>  | <b>CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)</b> |
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*Primary Examiner:* Miyoshi; Jesse Y

*Attorney, Agent or Firm:* Syncoda LLC

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of International Application No. PCT/CN2021/109103 filed on Jul. 29, 2021, which claims priority to Chinese Patent Application No. 202110356149.3 filed on Apr. 1, 2021. The disclosures of these applications are hereby incorporated by reference in their entirety.

### BACKGROUND

(1) A through-silicon via (TSV) technology is a technical solution which realizes interconnection

by stacking chips in a three-dimensional integrated circuit. The TSV technology becomes the most attractive technology in the current electronic packaging technologies as it can effectively stack chips in the three-dimensional direction, and therefore manufacture an electronic device with more complex structure, stronger performance and more cost-effective.

## SUMMARY

- (2) The application relates to the technical field of integrated circuits, and in particular to a semiconductor structure, a method for forming the same, and a stacked structure.
- (3) According to some embodiments, a semiconductor structure, a method for forming the same, and a stacked structure are provided.
- (4) A semiconductor structure includes a base, a connecting hole, a first insulating layer, a protective barrier layer, a second insulating layer and a connecting structure.
- (5) The base includes a substrate and a dielectric layer. The substrate is provided with a first front surface and a first back surface which are opposite. The dielectric layer is located at the first front surface.
- (6) The connecting hole penetrates through the substrate and extends into the dielectric layer.
- (7) The first insulating layer is located at the surface of an inner wall of the connecting hole.
- (8) The protective barrier layer is located at the surface of the first insulating layer. The protective barrier layer is grounded.
- (9) The second insulating layer is located at the surface of the protective barrier layer.
- (10) The connecting structure is located at the surface of the second insulating layer and filled in the connecting hole.
- (11) A method for forming a semiconductor structure includes the following operations.
- (12) A base is provided. The base includes a substrate and a dielectric base layer. The substrate is provided with a first front surface and a first back surface which are opposite. The dielectric base layer is located at the first front surface. A connecting hole is formed in the base. The connecting hole penetrates through the substrate and extends into the dielectric base layer.
- (13) A first insulating layer, a protective barrier layer, a second insulating layer and a connecting structure are formed in the connecting hole.
- (14) The first insulating layer is located at the surface of an inner wall of the connecting hole.
- (15) The protective barrier layer is located at the surface of the first insulating layer.
- (16) The second insulating layer is located at the surface of the protective barrier layer.
- (17) The connecting structure is located at the surface of the second insulating layer and filled in the connecting hole.
- (18) Part of the dielectric base layer is removed, so that the protective barrier layer is grounded. The retained dielectric base layer constitutes a dielectric layer.
- (19) A stacked structure is formed by processing any one of the above described semiconductor structure described in any one of the above.
- (20) The above description is only an overview of the technical solution of the application. In order to better understand the technical means of the application and implement it in accordance with the contents of the description, description in detail is performed below through better embodiments of the application and in combination with the drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In order to describe the technical solutions in the embodiments of the application or a conventional art more clearly, the drawings required to be used in descriptions about the embodiments or the conventional art will be simply introduced below. It is apparent that the figures described below are only some embodiments of the application. Other figures may further be

obtained by those of ordinary skilled in the art according to these drawings without creative work.

(2) FIG. 1 is a flowchart of a method for forming a semiconductor structure according to an embodiment.

(3) FIG. 2 is a flowchart of forming a first insulating layer, a protective barrier layer, a second insulating layer and a connecting structure in a connecting hole according to an embodiment.

(4) FIG. 3 is a flowchart of removing part of a connecting base line and part of a protective barrier base layer according to an embodiment.

(5) FIG. 4 is a flowchart of forming a second insulating layer at a surface of a protective barrier base layer according to an embodiment.

(6) FIG. 5 is a first structural schematic structural diagram showing a semiconductor structure manufacturing process.

(7) FIG. 6 is a second structural schematic structural diagram showing a semiconductor structure manufacturing process.

(8) FIG. 7 is a third structural schematic structural diagram showing a semiconductor structure manufacturing process.

(9) FIG. 8 is a fourth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(10) FIG. 9 is a fifth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(11) FIG. 10 is a sixth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(12) FIG. 11 is a seventh structural schematic structural diagram showing a semiconductor structure manufacturing process.

(13) FIG. 12 is an eighth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(14) FIG. 13 is a ninth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(15) FIG. 14 is a tenth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(16) FIG. 15 is an eleventh structural schematic structural diagram showing a semiconductor structure manufacturing process.

(17) FIG. 16 is a twelfth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(18) FIG. 17 is a thirteenth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(19) FIG. 18 is a fourteenth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(20) FIG. 19 is a fifteenth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(21) FIG. 20 is a sixteenth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(22) FIG. 21 is a seventeenth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(23) FIG. 22 is an eighteenth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(24) FIG. 23 is a nineteenth structural schematic structural diagram showing a semiconductor structure manufacturing process.

(25) FIG. 24 is a first schematic diagram of semiconductor structures in different embodiments.

(26) FIG. 25 is a second schematic diagram of semiconductor structures in different embodiments.

(27) FIG. 26 is a third schematic diagram of semiconductor structures in different embodiments.

(28) FIG. 27 is a fourth schematic diagram of semiconductor structures in different embodiments.  
(29) FIG. 28 is a first schematic diagram of processed semiconductor structures for stacking in different embodiments.  
(30) FIG. 29 is a second schematic diagram of processed semiconductor structures for stacking in different embodiments.

#### DETAILED DESCRIPTION

(31) In order to make the application convenient to understand, the application will be described more comprehensively below with reference to the related drawings. The drawings show embodiments of the application. However, the application may be implemented in various forms and is not limited to the embodiments described herein. On the contrary, these embodiments are provided to make the contents disclosed in the application understood more thoroughly and comprehensively.

(32) Unless otherwise defined, all technical and scientific terms used herein have the same meanings as commonly understood by a person skilled in the art that the application belongs to. Herein, terms used in the description of the application are only for the purpose of describing specific embodiments and not intended to limit the application.

(33) It is to be understood that description that an element or layer is “on”, “adjacent to”, “connected to”, or “coupled to” another element or layer may refer to that the element or layer is directly on, adjacent to, connected to or coupled to the other element or layer, or there may be an intermediate element or layer. On the contrary, description that an element is “directly on”, “directly adjacent to”, “directly connected to” or “directly coupled to” another element or layer refers to that there is no intermediate element or layer. It is to be understood, although various elements, components, regions, layers, doping types and/or parts may be described with terms first, second, third, etc., these elements, components, regions, layers, doping types and/or parts should not be limited by these terms. These terms are used only to distinguish one element, component, area, layer, doping type or part from another element, component, area, layer, doping type or part. Therefore, a first element, component, region, layer, doping type or part discussed below may be represented as a second element, component, region, layer or part without departing from the teaching of the disclosure.

(34) Spatially relational terms such as “below”, “under”, “lower”, “beneath”, “above”, and “upper” may be used herein for describing a relationship between one element or feature and another element or feature illustrated in the figures. It is to be understood that, in addition to the orientation shown in the figures, the spatially relational terms further include different orientations of devices in use and operation. For example, if the devices in the figures are turned over, elements or features described as being “under” or “beneath” or “below” other elements or features will be oriented to be “on” the other elements or features. Therefore, the exemplary terms “under” and “below” may include both upper and lower orientations. Moreover, the device may include otherwise orientation (such as rotation by 90 degrees or in other orientations) and the spatial descriptors used herein may be interpreted accordingly.

(35) As used herein, singular forms “a”, “an”, and “the/said” may also include plural forms, unless otherwise specified in the context. It is also to be understood that, terms such as “comprising/containing” or “having” represent the existence of declarative features, wholes, steps, operations, components, parts or combinations thereof, but not excluding the possibility of the existence or addition of one or more other features, wholes, steps, operations, components, parts or combinations of them. Meanwhile, in the specification, term “and/or” includes any and all combinations of the related listed items.

(36) Referring to a cross section diagram of a schematic diagram of an ideal embodiment (and an intermediate structure) of the disclosure herein, applied embodiments are described, so that change of shown shapes due to a manufacturing technology and/or tolerance may be predicted. Therefore, the embodiments of the disclosure should not be limited to specific shapes of shown areas, but

including shape deviation due to the manufacturing technology.

(37) Due to an electromagnetic field of transmission signals of TSV structures, the noise coupling between the TSV structures is quite serious. Particularly, in a high-density array of TSV structures, the influence of coupling capacitance and coupling inductance has become a major factor affecting the electrical performance of the TSV structure.

(38) In an embodiment, referring to FIG. 1, a method for forming a semiconductor structure includes the following operations.

(39) Referring to FIG. 5, at S100, a base 100 is provided. The base 100 includes a substrate 110 and a dielectric base layer 121. The substrate 110 is provided with a first front surface 110a and a first back surface 110b which are opposite. The dielectric base layer 121 is located at the first front surface 110a. A connecting hole 100a is formed in the base 100. The connecting hole 100a penetrates through the substrate 110 and extends into the dielectric base layer 121.

(40) Referring to FIG. 24, at S200, a first insulating layer 200, a protective barrier layer 300, a second insulating layer 400 and a connecting structure 500 are formed at the surface of an inner wall of the connecting hole 100a. The protective barrier layer 300 is grounded. The first insulating layer 200 is located at the surface of the inner wall of the connecting hole 100a. The protective barrier layer 300 is located at the surface of the first insulating layer 200. The second insulating layer 400 is located at the surface of the protective barrier layer 300. The connecting structure 500 is located at the surface of the second insulating layer 400 and filled in the connecting hole 100a. Part of the dielectric base layer 121 is removed, so that the protective barrier layer 300 is grounded. The retained dielectric base layer 121 constitutes a dielectric layer 120.

(41) In S100, the substrate 110 may include, but not limited to, a silicon substrate. A shallow trench isolation structure 600 may be formed at one side, close to the dielectric base layer 121, of the substrate 110. The shallow trench isolation structure 600 isolates the substrate 110 into multiple active areas. The active areas are configured to form various semiconductor devices.

(42) The dielectric base layer 121 may include, but not limited to, an oxide layer such as a silicon dioxide layer. An interconnecting hole structure and a metal layer electrically connected with the active area may be formed in the dielectric base layer 121, so as to lead out a signal of a semiconductor device or provide an external signal to the semiconductor device.

(43) The interior of the connecting hole 100a is configured to form a connecting structure 500. When chips formed by multiple semiconductor structures are stacked, the corresponding connecting holes 100a from various chips are aligned, so that the connecting structures 500 in the various corresponding connecting hole 100a are electrically connected, and thus interconnection among the various chips is realized.

(44) In S200, the first insulating layer 200 is located between the protective barrier layer 300 and the inner wall of the connecting hole 100a, so as to realize electrical isolation between the protective barrier layer 300 and the substrate 110. The material of the first insulating layer 200 may be silicon dioxide, etc. The material of the first insulating layer and the material of the dielectric base layer 121 may be the same or not.

(45) The material of the first insulating layer 200 may be silicon dioxide, etc.

(46) The material of the protective barrier layer 300 may be tantalum (Ta) and/or tantalum nitride (TaN), etc. Moreover, the protective barrier layer 300 is connected to the ground.

(47) The second insulating layer 400 is located between the connecting structure 500 and the protective barrier layer 300, so as to realize electrical isolation between the connecting structure 500 and the protective barrier layer 300. The material of the second insulating layer 400 may be silicon dioxide, etc. The material of the second insulating layer and the material of the first insulating layer 200 may be the same or not.

(48) The connecting structure 500 is a structure capable of performing conductive connection, and may include multiple film layers. Certainly, the connecting structure 500 may also only include one film layer, and there is no limitation in the application.

(49) After the connecting structure **500** is formed, annealing treatment is usually executed to make the size of particles such as metal in the connecting structure **500** more uniform, so as to reduce the resistivity and improve the electromigration resistance. However, after annealing, the grain size of the particles such as metal in the connecting structure **500** increases obviously, so that the structure surface formed by them will become rough, resulting in interface stress and increasing the possibility of current leakage.

(50) The material of the protective barrier layer **300** may be Ta and/or TaN, which can effectively inhibit the enlargement of the particles such as metal in the connecting structure **500** during annealing, so as to effectively reduce the interface stress. Meanwhile, the protective barrier layer **300** can also prevent the particles such as metal in the connecting structure **500** from diffusing to the substrate **110** and the dielectric layer **120**.

(51) The semiconductor structure usually includes multiple connecting structures **500**. Due to existence of an electromagnetic field of transmission signals of the connecting structures **500**, the various connecting structures **500** are easily affected by noise coupling. Particularly, in a high-density array of the connecting structures **500**, the connecting structures **500** are easily affected by coupling capacitance and coupling inductance, so as to affect the electrical performance of the connecting structures **500**.

(52) In the embodiment, on one hand, the protective barrier layer **300** located in the connecting hole **100a** is used to prevent the particles such as metal in the connecting structure **500** from becoming larger during annealing, so as to inhibit thermal stress. On the other hand, the protective barrier layer **300** located in the connecting hole **100a** is grounded, so that it also forms an electromagnetic shielding ring at the periphery of the connecting structure **500** while inhibiting stress, thereby effectively avoiding signal coupling among the various connecting structures **500** and among the connecting structures **500** and surrounding semiconductor devices.

(53) In an embodiment, referring to FIG. 5, the connecting hole **100a** is formed by etching from the first back surface **110b**.

(54) As an example, the base **100** as shown in FIG. 5 is etched from the first back surface **110b** through a dry etching method to form the connecting hole **100a**.

(55) The depth of the connecting hole **100a** may be 20-150  $\mu\text{m}$ , and the depth of the connecting hole **100** extending into the dielectric base layer **121** may be 0.5-1  $\mu\text{m}$ . The diameter of the connecting hole **100a** may be 3-50  $\mu\text{m}$ . The depth-to-width ratio (namely a ratio of the depth to the diameter) of the connecting hole **100a** may be 0.4-50.

(56) Since the first back surface **110b** of the substrate **110** is farther from the semiconductor device formed in the active area and a line structure connecting the semiconductor device, the embodiment can effectively prevent damage to the semiconductor device formed in the active area and the related circuit structure when forming the connecting hole **100a**.

(57) In an embodiment, referring to FIG. 18 and FIG. 25, a metal layer **700** is arranged in the dielectric base layer **121**. The protective barrier layer **300** is grounded through the metal layer **700**.

(58) Specifically, the metal layer **700** is used to lead out a signal in the semiconductor device or introduce an external signal.

(59) According to the application, the protective barrier layer **300** is grounded through the metal layer **700**, so that an electromagnetic shielding ring can be formed conveniently and effectively.

(60) In an embodiment, referring to FIG. 19, the dielectric base layer **122** is provided with a second front surface **120a**. The second front surface **120a** is far away from the first front surface **110a** of the substrate **110**.

(61) Referring to FIG. 2, S200 includes the following operations.

(62) At S210, referring to FIG. 6, a first insulating base layer **210** is formed at the surface of an inner wall of the connecting hole **100a**.

(63) At S220, referring to FIG. 7, a protective barrier base layer **310** is formed at the surface of the first insulating base layer **210**.

(64) At S230, referring to FIG. 14, the second insulating layer **400** is formed at the surface of the protective barrier base layer **310**.

(65) At S240, referring to FIG. 15, FIG. 16, FIG. 17 or FIG. 18, the connecting structure **500** is formed at the surface of the second insulating layer **400**.

(66) At S250, referring to FIG. 19, the dielectric base layer **122** and the first insulating base layer **210** are etched from the second front surface **120a** of the dielectric base layer **122** to form a trench **120c**. The trench **120c** exposes the protective barrier base layer **310** and the metal layer **700**. The retained dielectric base layer **122** constitutes the dielectric layer **120**, and the retained first insulating base layer **210** constitutes the first insulating layer **200**.

(67) At S260, referring to FIG. 20, a connecting base line **810** is formed in the trench **120c**.

(68) At S270, referring to FIG. 23, part of the connecting base line **810** and part of the protective barrier base layer **310** are removed. The retained connecting base line **810** constitutes a connecting line **800**, the retained protective barrier base layer **310** constitutes the protective barrier layer **300**. The connecting line **800** connects the protective barrier layer **300** to the metal layer **700**.

(69) In S210, a silicon dioxide film layer may be formed at the inner wall of the connecting hole **100a** as the first insulating base layer **210** by a chemical vapor deposition (CVD) method based on silane (SiH.sub.4) or tetraethyl orthosilicate (TEOS). The thickness of the first insulating base layer **210** may be 0.2-2  $\mu\text{m}$ .

(70) In S220, the protective barrier base layer **310** may be deposited and further formed by a physical vapor deposition (PVD) method. The material of the protective barrier base layer **310** may be Ta and/or TaN, etc. The thickness of the protective barrier base layer **310** may be 0.05-0.1  $\mu\text{m}$ .

(71) In S230, the material of the second insulating layer **400** and the material of the first insulating base layer **210** may be the same. Certainly, the material of the second insulating layer **400** and the material of the first insulating base layer **210** may also be different. There is no limitation to this in the application.

(72) In S240, the material of the connecting structure **500** may include metal, etc.

(73) In S250, the trench **120c** may be formed through dry etching. The material of the dielectric layer **120** and the material of the first insulating layer **200** may be the same.

(74) In S260, the connecting base line **810** may be formed through PVD and the like. The connecting base line **810** covers the dielectric layer **120** at the inner wall of the trench **120c**, the metal layer **700** and the protective barrier base layer **310**.

(75) In S270, the protective barrier layer **300** and the metal layer **700** are effectively connected through the connecting line **800**.

(76) In the embodiment, the protective barrier layer **300** and the metal layer **700** are connected through the connecting line **800**, so that the protective barrier layer **300** can be grounded through the metal layer **700**.

(77) Certainly, in other embodiments, the protective barrier layer **300** may be grounded through the metal layer **700** in other ways, For example, the dielectric base layer **122** and the first insulating base layer **210** may also be etched from the second front surface **120a** of the dielectric base layer **122** to form the trench **120c**, and then the metal layer for leading out a signal is formed in the trench **120c**, so that the protective barrier layer **300** is directly connected with the metal layer and grounded through the metal layer.

(78) In an embodiment, referring to FIG. 3, S270 may include the following operations.

(79) At S271, referring to FIG. 21, a first insulating initial structure **911** is formed at the surface of the connecting base line **810**. The first insulating initial structure **911** fills up the trench **120c**.

(80) At S272, referring to FIG. 22, part of the first insulating initial structure **911** is removed so as to expose part of the connecting base line **810**. The retained first insulating initial structure **911** constitutes a first insulating part **910**. The distance between the first insulating part **910** and the second front surface **120a** is greater than the distance between the protective barrier layer **300** and the second front surface **120a**.



(81) At S273, referring to FIG. 23, the connecting base line **810** and the protective barrier base layer **310** exposed between the first insulating part **910** and the second front surface **120a** are removed. The retained connecting base line **810** constitutes the connecting line **800**, and the retained protective barrier base layer **310** constitutes the protective barrier layer **300**.

(82) At S274, referring to FIG. 24, a second insulating part **920** is formed in the trench **120c**, and the second insulating part **920** and the first insulating part **910** constitute an insulating structure **900**.

(83) In S271, a first insulating initial structure material (such as a silicon dioxide material) may be deposited on the surface of the connecting base line **810**, and then polished to form the first insulating initial structure **911**.

(84) In S272, after part of the first insulating initial structure **911** is removed, the retained first insulating initial structure **911** constitutes the first insulating part **910**.

(85) At this time, the exposed part of the connecting base line **810** is a part that needs to be removed to form the connecting line **800**.

(86) In S273, the connecting base line **810** and the protective barrier base layer **310** between the first insulating part **910** and the second front surface **120a** may be removed through a wet etching method. At this time, a second sub insulating layer **420** is exposed.

(87) In S274, a second insulating initial structure material (such as a silicon dioxide material) may be deposited in the trench **120c**, and then polished to form the second insulating part **920**. The second insulating part **920** and the first insulating part **910** constitute the insulating structure **900**.

(88) In the embodiment, the connecting line **800** and the protective barrier layer **300** are insulated and closed through the insulating structure **900**, so that the protective barrier layer **300** is stably grounded. At this time, the protective barrier layer **300**, the connecting line **800** and the metal layer **700** form an electromagnetic shielding ring together, so as to prevent interference of the connecting structure **500** to peripheral signals.

(89) Meanwhile, in the embodiment, the insulating structure **900** insulating and closing the connecting line **800** and the protective barrier layer **300** includes the second insulating part **920** and the first insulating part **910**.

(90) Certainly, in other embodiments, it may also form a structure with other forms by other methods. For example, after S273, the first insulating part **910** may also be removed, and then deposited and polished again to form an insulating structure.

(91) In an embodiment, referring to FIG. 24, the second insulating layer **400** includes a first sub insulating layer **410** and a second sub insulating layer **420** spaced apart from each other. The protective barrier layer **300**, the first sub insulating layer **410**, the connecting structure **500** and the second sub insulating layer **420** are arranged in a surrounding manner to form an air gap **100b**. Due to the poor thermal conductivity of air, the air gap **100b** formed in the embodiment can effectively prevent heat generated by the connecting structure **500** from diffusing to the surrounding semiconductor device or dielectric layer, etc.

(92) Moreover, at this time, the air gap **100b** can also effectively isolate the connecting structure **500** from the substrate **110** and/or the dielectric layer **120** around it, so as to effectively prevent the stress deformation of the substrate **110** and/or the dielectric layer **120** due to the thermal expansion of the connecting structure **500**, and then effectively prevent the crack phenomenon at the interface of the substrate **110** and/or the dielectric layer **120**.

(93) In an embodiment, referring to FIG. 24, the air gap **100b** passes through an interface between the substrate **110** and the dielectric layer **120**.

(94) The interface between the substrate **110** and the dielectric layer **120** is easily affected by the thermal expansion of the connecting structure **500** to cause stress deformation, resulting in interface cracks.

(95) In the embodiment, the air gap **100b** passes through the interface between the substrate **110** and the dielectric layer **120**, so that the air gap **100b** isolates the interface between the substrate **110**

and the dielectric layer **120** from the connecting structure **500**, and thus, the cracks at the interface of the interface between can be effectively prevented.

(96) Meanwhile, semiconductor devices are dense around the interface between the substrate **110** and the dielectric layer **120**. The structure of the semiconductor devices usually penetrates through both sides of the interface between the substrate **110** and the dielectric layer **120**.

(97) The air gap **100b** passes through the interface between the substrate **110** and the dielectric layer **120**, so that the air gap **100b** simultaneously isolates the connecting structure **500** from the substrate **110** and the dielectric layer **120** on both sides of the interface. Therefore, the embodiment can also effectively protect the semiconductor devices around the air gap **100b** from thermal insulation, and can also prevent the semiconductor devices around the air gap **100b** from affecting the performance due to stress deformation of the substrate **110** and/or the dielectric layer **120**.

(98) In an embodiment, the height of the portion of the air gap **100b** corresponding to the dielectric layer **120** is less than the height of the portion corresponding to the substrate **110**.

(99) The dielectric layer **120** is a film layer formed on the substrate **110**, and its thickness is much smaller than that of the substrate **110**. Therefore, in the embodiment, the height of the portion of the air gap **100b** corresponding to the dielectric layer **120** is relatively small, so that the structural instability of the dielectric layer **120** caused by the air gap **100b** can be effectively prevented.

(100) Moreover, heat generated in the dielectric layer **120** is relatively large. At this time, the height of the air gap **100b** corresponding to the substrate **110** is relatively large, which makes the heat flow to the position with large gap, and thus is more conducive to heat dissipation and prevent the influence on peripheral devices.

(101) In an embodiment, referring to FIG. 4, S230 includes the following operations.

(102) At S231, referring to FIG. 8, a primary insulating layer **411** is formed at the surface of the protective barrier base layer **310**. The primary insulating layer **411** includes a side wall part **4111**.

(103) At S232, referring to FIG. 9, a primary barrier layer **5111** is formed at the surface of the primary insulating layer **411**.

(104) At S233, referring to FIG. 10, a primary conductive layer **5211** is formed at the surface of the primary barrier layer **5111**.

(105) At S234, referring to FIG. 11, part of the primary conductive layer **5211** and part of the primary barrier layer **5111** are removed to expose part of the side wall part **4111**. The retained primary conductive layer **5211** constitutes a first conductive layer **521**, and the retained primary barrier layer **5111** constitutes a first sub barrier layer **511**.

(106) At S235, referring to FIG. 12, part of the side wall part **4111** is removed. The retained primary insulating layer **411** constitutes a first sub insulating layer **410**. The distance H1 between the first sub insulating layer **410** and an opening of the connecting hole **100a** is greater than the distance H2 between the first sub barrier layer **511** and the opening of the connecting hole **100a**.

(107) At S236, referring to FIG. 13, a second sub insulating layer **420** is formed at the surface of the protective barrier base layer **310**. The second sub insulating layer **420** and the first sub insulating layer **410** constitute the second insulating layer **400**.

(108) At this time, the S240 includes the following operations.

(109) At S241, referring to FIG. 14, a second sub barrier layer **512** is formed at the surface of the second sub insulating layer **420**, the surface of the first sub barrier layer **511** and the surface of the first conductive layer **521**. The second sub barrier layer **512** and the first sub barrier layer **511** constitute a barrier layer **510**.

(110) At S242, referring to FIG. 15, a second conductive layer **522** is formed at the surface of the second sub barrier layer **512**. The second conductive layer **522** and the first conductive layer **521** constitute a conductive structure **520**. The conductive structure **520** and the barrier layer **510** constitute the connecting structure **500**.

(111) In S231, the material of the primary insulating layer **411** may be insulating materials, such as silicon dioxide.

(112) As an example, when the material of the primary insulating layer **411** is silicon dioxide, a silicon dioxide film layer with the thickness of 0.2-2  $\mu\text{m}$  may be deposited at the inner wall of the connecting hole **100a** as the primary insulating layer **411** by a CVD method based on SiH.sub.4 or TEOS.

(113) In **S232**, the primary barrier layer **5111** may also be formed by a PVD method. The material of the primary barrier layer **5111** may be Ta and/or TaN. The thickness of the primary barrier layer **5111** may be 0.05-0.1  $\mu\text{m}$ .

(114) In **S233**, the material of the primary conductive layer **5211** may be metallic copper (Cu) and the like. Specifically, a copper seed layer may be formed on the surface of the primary conductive layer **5211** through the PVD method. Afterwards, electroplated Cu is formed on the surface of the copper seed layer through electroplating growth. The copper seed layer and the electroplated Cu together constitute the primary conductive layer **5211**.

(115) More specifically, since part of the primary conductive layer **5211** is also required to be removed in the subsequent **S234** to form the air gap **100b**, at this time, the primary conductive layer **5211** growing on the surface of the primary barrier layer **5111** may not fill up the connecting hole **100a**. As an example, the filling thickness of the central portion of the primary conductive layer **5211** may be 20%-70% of the depth of the connecting hole **100a**.

(116) In **S234**, part of the primary barrier layer **5111** and part of the primary conductive layer **5211** which cover the side wall part **4111** may be removed by a mixed acid solution, so that the side wall part **4111** of the primary insulating layer **411** is partially exposed, and the first sub barrier layer **511** and the first conductive layer **521** are formed at the same time.

(117) The first sub barrier layer **511** and the subsequently formed second sub barrier layer **512** constitute a barrier layer, which inhibits the thermal expansion stress generated by the conductive structure **520**. Therefore, the materials of the first sub barrier layer **511** and the second sub barrier layer **512** are materials with the function of blocking thermal stress such as Ta or TaN.

(118) After the side wall part **4111** of the primary insulating layer **411** is partially exposed, **S550** is executed, so that the air gap **100b** can be realized more easily.

(119) In **S235**, the side wall part **4111** of the primary insulating layer **411** (such as a silicon dioxide film layer) in the connecting hole **100a** may be etched by hydrofluoric acid or dry etching to form the first sub insulating layer **410**.

(120) Specifically, after part of the side wall part **4111** is removed, the distance H1 between the surface of the first sub insulating layer **410** formed by the retained primary insulating layer **411** and an opening of the connecting hole **100a** is greater than the distance H2 between the surface of the first sub barrier layer **511** and the opening of the connecting hole **100a**. That is, after part of the side wall part **4111** is removed in the direction as shown in FIG. 11, the surface of the first sub insulating layer **410** formed by the retained primary insulating layer **411** is lower than the surface of the first sub barrier layer **511**.

(121) In **S236**, specifically, the second sub insulating layer **420** may be formed through the CVD method. The material of the second sub insulating layer **420** and the material of the first sub insulating layer **410** may be the same or not.

(122) In **S241**, the material of the second sub barrier layer **512** and the material of the first sub barrier layer **511** may be the same. Certainly, they may also be different.

(123) Meanwhile, since the second sub barrier layer **512** is formed at the surface of the second sub insulating layer **420**, the surface of the first sub barrier layer **511** and the surface of the first conductive layer **521**, the second sub barrier layer **512** is connected with the first sub barrier layer **511**, thereby well inhibiting the thermal expansion stress of the conductive structure **330** therein.

(124) In **S242**, the material of the second conductive layer **522** and the material of the first conductive layer **521** may be the same or not, which is not limited in the application.

(125) As an example, when both the material of the second conductive layer **522** and the material of the first conductive layer **521** are Cu, the formation process may be similar to that of the primary

conductive layer **5211**. A copper seed layer may be formed, and then electroplated Cu is formed on the Cu seed crystal layer to form the second conductive layer **522**.

(126) In the embodiment, the primary insulating layer **411** is formed, and then part of the side wall part **4111** thereof is removed, so that an air gap **100b** can be formed conveniently and effectively.

(127) Moreover, in the embodiment, the second sub barrier layer **512** is provided between the second conductive layer **522** and the first conductive layer **521**. Therefore, the second sub barrier layer **512** can effectively prevent the possible inconsistency between the second conductive layer **522** and the first conductive layer **521** which are formed in two phases due to grain size, etc., and cracks at the interface due to the influence of thermal stress in the subsequent annealing process.

(128) Certainly, the application is not limited to this. In other embodiments, in **S241**, a second sub barrier base layer is formed at the surface of the second sub insulating layer **420**, the surface of the first sub barrier layer **511** and the surface of the first conductive layer **521**.

(129) Afterwards, at least part of the second sub barrier base layer at the surface of the first conductive layer **521** may be removed at first. The retained second sub barrier base layer constitutes the second sub barrier layer **512**. The second sub barrier layer **512** and the first sub barrier layer **511** constitute the barrier layer **520**.

(130) Then, referring to FIG. **16** or FIG. **17**, the second conductive layer **522** is formed at the surface of the second sub barrier layer **512** and the exposed surface of the first conductive layer **521**.

(131) At this time, the second conductive layer **522** and the first conductive layer **521** are contacted and connected, thereby reducing the contact resistance between the second conductive layer **522** and the first conductive layer **521**.

(132) Referring to FIG. **17**, when the second sub barrier layer **512** covering the surface of the first conductive layer **521** is partially removed, not only the contact resistance is reduced but also the influence of thermal stress between the second conductive layer **522** and the first conductive layer **521** which are formed in two phases is prevented.

(133) Meanwhile, when the central portion of the second sub barrier layer **512** covering the surface of the first conductive layer **521** is removed and the edge portion retains, the retained edge portion can also help to prevent the poor contact between the second sub barrier layer **512** and the first sub barrier layer **511** from affecting the inhibition of the thermal expansion stress of the conductive structure **520** (including the first conductive layer **521** and the second conductive layer **522**) therein.

(134) In an embodiment, referring to FIG. **15**, FIG. **16** or FIG. **17**, the thickness of the second sub barrier layer **512** is greater than that of the first sub barrier layer **511**.

(135) At this time, the poor contact between the second sub barrier layer **512** and the first sub barrier layer **511** may be effectively prevented from affecting the inhibition of the thermal expansion stress of the conductive structure **520** (including the first conductive layer **521** and the second conductive layer **522**) therein.

(136) It should be understood that although various steps in the flowcharts of FIG. **1** to FIG. **4** are sequentially displayed by following arrows, the steps are not sequentially executed necessarily in the order indicated by the arrows. Unless explicitly indicated in the disclosure, there is no strict sequence restriction on the execution of these steps, and these steps may be executed in other order. Moreover, at least part of steps in FIG. **1** to FIG. **4** may include multiple steps or multiple stages, these steps or stages are not executed necessarily at the same time, and may be executed at different times. These steps or stages are not sequentially executed necessarily, and may be executed in turn or alternatively with other steps or at least part of steps or stages in other steps.

(137) In an embodiment, a semiconductor structure is further provided. Referring to FIG. **25**, the semiconductor structure may include a base **100**, a connecting hole **100a**, a first insulating layer **200**, a protective barrier layer **300**, a second insulating layer **400** and a connecting structure **500**.

(138) The base **100** includes a substrate **110** and a dielectric layer **120**. The substrate **110** is

provided with a first front surface **110a** and a first back surface **110b** which are opposite. The dielectric layer **120** is located at the first front surface **110a**. The connecting hole **100a** penetrates through the substrate **110** and extends into the dielectric layer **120**. The first insulating layer **200** is located at the surface of the inner wall of the connecting hole **100a**. The protective barrier layer **300** is located at the surface of the first insulating layer **200** and grounded. The second insulating layer **400** is located at the surface of the protective barrier layer **300**. The connecting structure **500** is located at the surface of the second insulating layer **400** and filled in the connecting hole **100a**.

(139) In the embodiment, on one hand, the protective barrier layer **300** located in the connecting hole **100a** is configured to prevent the particles such as metal in the connecting structure **500** from becoming larger during annealing, so as to inhibit thermal stress. On the other hand, the protective barrier layer **300** located in the connecting hole **100a** is grounded, so that it also forms an electromagnetic shielding ring on the periphery of the connecting structure **500** while inhibiting stress, so as to effectively avoid signal coupling among the various connecting structures **500** and among the connecting structures **500** and surrounding semiconductor devices.

(140) In an embodiment, continuously referring to FIG. 25, a metal layer is arranged in the dielectric layer **120**. The protective barrier layer **300** is grounded through the metal layer.

(141) In an embodiment, continuously referring to FIG. 25, a connecting line **800** is also arranged in the dielectric layer **120**. The connecting line **800** connects the protective barrier layer **300** with the metal layer **700**. The metal layer **700**, the connecting line **800** and the protective barrier layer **300** together constitute an electromagnetic shielding ring.

(142) In an embodiment, continuously referring to FIG. 25, an insulating structure **900** is also arranged in the dielectric layer **120**. The insulating structure **900** insulates and closes the connecting line **800** and the protective barrier layer **300**.

(143) In an embodiment, continuously referring to FIG. 25, the insulating structure **900** includes a first insulating part **910** and a second insulating part **920**. The first insulating part **910** is located between the connecting line **800** and the second insulating part **920**. The second insulating part **920** insulates and closes the connecting line **800** and the protective barrier layer **300**.

(144) In an embodiment, referring to FIG. 24, FIG. 26 or FIG. 27, the second insulating layer **400** includes a first sub insulating layer **410** and a second sub insulating layer **420** spaced apart from each other. The protective barrier layer **300**, the first sub insulating layer **410**, the connecting structure **500** and the second sub insulating layer **420** are arranged in a surrounding manner to form an air gap **100b**.

(145) In an embodiment, referring to FIG. 24, the air gap **100b** passes through an interface between the substrate **110** and the dielectric layer **120**.

(146) In an embodiment, the height of the portion of the air gap **100b** corresponding to the dielectric layer **120** is less than the height of the portion corresponding to the substrate **110**.

(147) In an embodiment, referring to FIG. 26 or FIG. 27, the connecting structure **500** includes a barrier layer **510** and a conductive structure **520**. The barrier layer **510** includes a first sub barrier layer **511** and a second sub barrier layer **512** which are interconnected. The conductive structure **520** includes a first conductive layer **521** and a second conductive layer **522**. The first sub barrier layer **511** is located at the surface of the first sub insulating layer **410**, and the first conductive layer **521** is located at the surface of the first sub barrier layer **511**. The second sub barrier layer **512** is located at the surface of the second sub insulating layer **420**. The second conductive layer **522** is located at the surface of the second sub barrier layer **512**, and is filled in the connecting hole **100a**.

(148) The second conductive layer **522** and the first conductive layer **521** are separated by the second sub barrier layer **512** (referring to FIG. 27). Alternatively, the second conductive layer **522** is connected with the first conductive layer **521** (referring to FIG. 26).

(149) In an embodiment, referring to FIG. 27, the thickness of the second sub barrier layer **512** is greater than that of the first sub barrier layer **511**.

(150) In an embodiment, a stacked structure is further provided. The stacked structure is formed by

processing any one of the above-described semiconductor structure.

(151) Specifically, when the stacked structure is formed, the above semiconductor structure needs to be subjected to processing technologies such as etching or flattening to expose the connecting structure **500** in the dielectric layer **120** (referring to FIG. **28** or FIG. **29**). By doing so, the connecting structure **500** in the connecting hole **100a** is electrically connectable when it is stacked with adjacent semiconductor structures, thus can be used for signal transmission.

(152) More specifically, when the connecting structure **500** includes the barrier layer **510** and the conductive structure **520**, the conductive structure **520** needs to be exposed for signal transmission.

(153) The specific definition and technical effect of the semiconductor structure may refer to the above definition of the formation method of the semiconductor structure, which will not be elaborated here.

(154) Certainly, it should be understood that, the semiconductor structure of the application is not limited to the formation method of the semiconductor structure in the above embodiment.

(155) In the descriptions of the specification, the descriptions made with reference to terms “an embodiment”, “an ideal embodiment” or the like refer to that specific features, structures, materials or characteristics described in combination with the embodiment or the example are included in at least one embodiment or example of the disclosure. In the specification, schematic description on the above terms not always refers to same embodiments or demonstrations.

(156) Each technical feature of the above mentioned embodiments may be combined freely. For simplicity of description, not all possible combinations of each technical solution in the above mentioned embodiments are described. However, any combination of these technical features shall fall within the scope recorded in the specification without conflicting.

(157) The above mentioned embodiments only express some implementation modes of the application and are specifically described in detail and not thus understood as limits to the patent scope of the application. It is to be pointed out that those of ordinary skill in the art may further make multiple transformations and improvements without departing from the concept of the application and all of these shall fall within the scope of protection of the application. Therefore, the scope of patent protection of the application should be subject to the appended claims.

## Claims

1. A semiconductor structure, comprising: a base, comprising a substrate and a dielectric layer, the substrate being provided with a first front surface and a first back surface which are opposite, and the dielectric layer being located at the first front surface; a connecting hole, penetrating through the substrate and extending into the dielectric layer; a first insulating layer, located at a surface of an inner wall of the connecting hole; a protective barrier layer, located at a surface of the first insulating layer and grounded; a second insulating layer, located at a surface of the protective barrier layer, the second insulating layer comprises a first sub insulating layer and a second sub insulating layer spaced apart from each other; and a connecting structure, located at a surface of the second insulating layer and filling up the connecting hole; and wherein the protective barrier layer, the first sub insulating layer, the connecting structure and the second sub insulating layer are arranged in a surrounding manner to form an air gap.

2. The semiconductor structure according to claim 1, wherein a metal layer is arranged in the dielectric layer, and the protective barrier layer is grounded through the metal layer.

3. The semiconductor structure according to claim 2, wherein a connecting line is also arranged in the dielectric layer, the connecting line connects the protective barrier layer with the metal layer, and wherein the metal layer, the connecting line and the protective barrier layer together constitute an electromagnetic shielding ring.

4. The semiconductor structure according to claim 3, wherein an insulating structure is also arranged in the dielectric layer, and the insulating structure seals the connecting line and the

protective barrier layer.

5. The semiconductor structure according to claim 4, wherein the insulating structure comprises a first insulating part and a second insulating part, the first insulating part is located between the connecting line and the second insulating part, and the second insulating part seals the connecting line and the protective barrier layer.

6. The semiconductor structure according to claim 1, wherein the air gap passes through an interface between the substrate and the dielectric layer.

7. The semiconductor structure according to claim 6, wherein a height of a portion of the air gap corresponding to the dielectric layer is less than a height of a portion of the air gap corresponding to the substrate.

8. The semiconductor structure according to claim 1, wherein the connecting structure comprises a barrier layer and a conductive structure, the barrier layer comprises a first sub barrier layer and a second sub barrier layer which are interconnected, the conductive structure comprises a first conductive layer and a second conductive layer, the first sub barrier layer is located at a surface of the first sub insulating layer, the first conductive layer is located at a surface of the first sub barrier layer, the second sub barrier layer is located at a surface of the second sub insulating layer, the second conductive layer is located at a surface of the second sub barrier layer and filled in the connecting hole, the second conductive layer and the first conductive layer are separated by the second sub barrier layer, or the second conductive layer is connected with the first conductive layer.

9. The semiconductor structure according to claim 8, wherein a thickness of the second sub barrier layer is greater than a thickness of the first sub barrier layer.

10. A stacked structure, formed by processing the semiconductor structure according to claim 1.

11. A method for forming a semiconductor structure, comprising: providing a base, wherein the base comprises a substrate and a dielectric base layer, the substrate is provided with a first front surface and a first back surface which are opposite, the dielectric base layer is located at the first front surface, a connecting hole is formed in the base, and the connecting hole penetrates through the substrate and extends into the dielectric base layer; and forming a first insulating layer, a protective barrier layer, a second insulating layer and a connecting structure in the connecting hole, wherein the first insulating layer is located at a surface of an inner wall of the connecting hole, the protective barrier layer is located at a surface of the first insulating layer, the second insulating layer is located at a surface of the protective barrier layer, and the connecting structure is located at a surface of the second insulating layer and filled in the connecting hole, wherein the second insulating layer comprises a first sub insulating layer and a second sub insulating layer spaced apart from each other; wherein part of the dielectric base layer is removed, so that the protective barrier layer is grounded, and the dielectric base layer that is retained constitutes a dielectric layer, and the protective barrier layer, the first sub insulating layer, the connecting structure and the second sub insulating layer are arranged in a surrounding manner to form an air gap.

12. The method according to claim 11, wherein the connecting hole is formed by etching from the first back surface.

13. The method according to claim 11, wherein the dielectric base layer is provided with a metal layer, and the protective barrier layer is grounded through the metal layer.

14. The method according to claim 13, wherein the dielectric base layer is provided with a second front surface which is far away from the first front surface; said forming a first insulating layer, a protective barrier layer, a second insulating layer and a connecting structure in the connecting hole comprising: forming a first insulating base layer at the surface of the inner wall of the connecting hole; forming a protective barrier base layer at a surface of the first insulating base layer; forming the second insulating layer at a surface of the protective barrier base layer; forming the connecting structure at the surface of the second insulating layer; etching the dielectric base layer and the first insulating base layer from the second front surface of the dielectric base layer to form a trench, the trench exposing the protective barrier base layer and the metal layer, the dielectric base layer that is

retained constituting the dielectric layer, and the first insulating base layer that is retained constituting the first insulating layer; forming a connecting base line in the trench; and removing part of the connecting base line and part of the protective barrier base layer, the connecting base line that is retained constituting a connecting line, the protective barrier base layer that is retained constituting the protective barrier layer, and the connecting line connecting the protective barrier layer with the metal layer.

15. The method according to claim 14, said removing part of the connecting base line and part of the protective barrier base layer comprising: forming a first insulating initial structure at a surface of the connecting base line, the first insulating initial structure filling up the trench; removing part of the first insulating initial structure so as to expose part of the connecting base line, the first insulating initial structure that is retained constituting a first insulating part, and a distance between the first insulating part and the second front surface is greater than a distance between the connecting structure and the second front surface; removing the connecting base line and the protective barrier base layer between the first insulating part and the second front surface, the connecting base line that is retained constituting the connecting line, and the protective barrier base layer that is retained constituting the protective barrier layer; and forming a second insulating part in the trench, the second insulating part and the first insulating part constituting an insulating structure.

16. The method according to claim 11, wherein the air gap passes through an interface between the substrate and the dielectric layer.

17. The method according to claim 11, said forming the second insulating layer at a surface of the protective barrier base layer comprising: forming a primary insulating layer at the surface of the protective barrier base layer, the primary insulating layer comprising a side wall part; forming a primary barrier layer at a surface of the primary insulating layer; forming a primary conductive layer at a surface of the primary barrier layer; removing part of the primary conductive layer and part of the primary barrier layer so as to expose part of the side wall part, the primary conductive layer that is retained constituting a first conductive layer, and the primary barrier layer that is retained constituting the first sub barrier layer; removing part of the side wall part, the primary insulating layer that is retained constituting a first sub insulating layer, and a distance between the first sub insulating layer and an opening of the connecting hole being greater than a distance between the first sub barrier layer and the opening of the connecting hole; and forming a second sub insulating layer at the surface of the protective barrier base layer, the second sub insulating layer and the first sub insulating layer constituting the second insulating layer, said forming the connecting structure at the surface of the second insulating layer comprising: forming a second sub barrier layer at a surface of the second sub insulating layer, a surface of the first sub barrier layer and a surface of the first conductive layer, the second sub barrier layer and the first sub barrier layer constituting a barrier layer; and forming a second conductive layer on the surface of the second sub barrier layer, the second conductive layer and the first conductive layer constituting the conductive structure, and the conductive structure and the barrier layer constituting the connecting structure.

18. The method according to claim 17, wherein a thickness of the second sub barrier layer is greater than a thickness of the first sub barrier layer.

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